A family of defining quartics in 226 variables

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Abstract

Under the specified modular-realization, elliptic, pole-realization and prime-pattern inputs, we give an integer quartic family in 226 total variables whose sufficiently large members define the integers in the rationals. The proved contact-height exponent 7/3 becomes 7/6 on two disjoint contact blocks. A common elliptic chart uses four field coordinates, and all elliptic comparison ratios use the plain formal parameter. Five factors represent sixteen local tests. Clearing each factor equation by an inverse fourth power removes five separately stored field fourth powers. Three reciprocal weights per detector suffice, and the local comparison order is three. An inverse weight also gives a contact circuit with three scalar outputs per detector. The quartic is a sum of 200 primitive quadratic squares, has 2835 monomials, and has coefficient height $149^{4N+2}+1$ for $N \geq 53$. The sufficient defining parameter remains ineffective because of the additive constant in the aggregate trace-height theorem. No additional *abc* conjecture is used.

1 The family and its arithmetic scope

All variable counts include the distinguished input X. For a fixed integer $N \geq 1$ let $A_N = 149^{2N+1}$. The complete tables

`quadratic-family226-v21.json`
`polynomial-family226-v21.json`

specify 200 primitive integer quadratic polynomials and their sum of squares

$$P_N(X, Y_1, \ldots, Y_{225}) = \sum_{j=1}^{200} Q_{N,j}^2.$$

The parameter A_N is a coefficient, not a variable.

Theorem 1.1. *Assume the arithmetic premises stated below. There is a fixed integer $N_0 \geq 53$ such that every $N \geq N_0$ satisfies*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{225}\ P_N(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

For every $N \geq 1$ the polynomial P_N has degree exactly four, content one, and 2835 nonzero coordinate monomials. For $N \geq 53$,

$$H(P_N) = 149^{4N+2} + 1.$$

The construction specifies every member exactly. The proof does not determine a sufficient numerical value of N_0.

Set

$$K=\mathbb{Q}(\rho),\quad \rho^2=2,\quad D=(3-\rho)^2=11-6\rho,\qquad E:y^2=x^3-Dx,$$
$$P=2(1+\rho,2\rho)=\left(\frac{57-28\rho}{8},\frac{-660+417\rho}{32}\right).$$

We retain the following arithmetic premises, with the precise domains and hypotheses recorded in [1, 2, 3]:

1. $2E(K)=\mathbb{Z}P$ and $h(x([n]P))\geq n^2-6$ for every nonzero integer n, together with the specified point charts and group domains.

2. The modular-realization input of [7, Definition 6.1], with its normalized coefficient character. The multiplicity theorem of [8, Theorem 4.1] proves

$$h(u)\leq HM_5(u)^{7/3},\qquad H=2^{221234}. \tag{1}$$

   Here $M_5(u)$ is the odd contact radical outside 76176 for $\infty,0,1,2401,-343/32$, and u avoids the four finite contacts. Explicitly, if $u=r/s$ is reduced with $s>0$, then $M_5(u)$ is the product of primes $p>76176$ having a positive odd valuation on at least one of

$$s,\quad r,\quad r-s,\quad r-2401s,\quad 32r+343s.$$

   The estimate is a consequence of the stated modular input, rather than an additional *abc* assumption.

3. The pole obstruction outside $2,3,17$, and its simultaneous realization for positive inputs less than 17 with square reduced denominator and the stated local squareclasses and valuation bounds at those three primes. Its normalized quadratic equations are retained.

4. The simultaneous prime-pattern, conjugate reduction-order, finite congruence, and real approximation assertions used by the source. The weighted selection and fixed denominator dilation are justified from the cited analytic estimates in [10].

The aggregate trace-height theorem used below is the proved consequence of the family height theorem described in [6]; its additive constant is ineffective. The contact amplification below is an elementary consequence of (1), not an additional arithmetic assumption. Exact coefficient verification does not prove the retained arithmetic premises.

2 Two contact blocks and the comparison scales

Put

$$\Theta=\{0,-1/2,-1,1/2399,-32/407\},\qquad \Theta_2=\Theta\cup(2-\Theta),\qquad F_2(t)=\prod_{\theta\in\Theta_2}(t-\theta).$$

The ten contacts are distinct, omit one, and are symmetric about one. Let Σ consist of the primes at most 76176 and the prime 2029147. Outside this set, all contact differences and all numbers $1-\theta$ are units. Define $M_2^\circ(t)$ as the product of primes $p\notin\Sigma$ for which $v_p(t-\theta)$ is positive and odd for some $\theta\in\Theta_2$. Exact factorizations in the independent certificate verify this exceptional set. In particular

$$F_2(1)=-\frac{82429855236}{7878870169},\qquad M_2^\circ(1)=1.$$

Set $E_2=2029147$ and $H_2=2^{221284}$. Then

$$h(t)\leq H_2M_2^\circ(t)^{7/6}\qquad (t\notin\Theta_2). \tag{2}$$

Indeed, apply (1) to $2 + 1/t$ and $2 + 1/(2 - t)$. The inverse projective matrices have maximum absolute row sums three and seven, respectively. If M_+, M_- are the two block radicals with Σ omitted, the original radicals are at most $E_2 M_+$ and $E_2 M_-$. Outside Σ these radicals are coprime, with $M_+ M_- = M_2^\circ(t)$. Multiplying the two inequalities

$$h(t) - \log 7 \le H E_2^{7/3} M_\pm^{7/3}$$

when the left side is positive gives the exponent $7/6$. If it is nonpositive, the asserted bound is immediate. Finally $E_2 < 2^{21}$ and $\log 7 < 2$ give

$$\log 7 + H E_2^{7/3} < 2 + 2^{221283} < H_2.$$

The retained root points are

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$

with nonzero integer labels. Set

$$T = \frac{1}{4} \operatorname{Tr}_{K/\mathbb{Q}} \left(\sum_{i=1}^{4} x(Q_i) \right).$$

It is positive, and the aggregate trace theorem gives fixed constants $M_1 > 0$, $M_2 \ge 1$ such that

$$B^2 \le M_1 h(T) + M_2. \tag{3}$$

Use the two positive rational scales $a_* = 1, T$. No square condition on a scale is needed. Replacing the half-trace in the aggregate height theorem by its half changes its logarithmic height by at most $\log 2$; this is absorbed into M_2. For each scale put $t_* = a_* \lambda$ and $\delta = t_* - 1$. On the nonzero domain use the balanced contact and the local weight

$$R = \frac{F_2(t_*)}{\delta^{10}}, \qquad q = \frac{\delta^{10}}{F_2(t_*)} - 1 = R^{-1} - 1, \qquad \gamma = \frac{408}{409}. \tag{4}$$

Let e_j be the jth elementary symmetric sum of the five positive rational numbers $(1 - \theta)^2$, $\theta \in \Theta$. Thus

$$P_5(r) = \prod_{\theta \in \Theta} (r - (1 - \theta)^2) = r^5 - e_1 r^4 + e_2 r^3 - e_3 r^2 + e_4 r - e_5, \qquad e_5 \ne 0.$$

Use the inverse weight s_1 required by the local factors, with $q s_1 = 1$. The contact equations retain exactly three scalar outputs r, r_2, v:

$$\begin{aligned} r &= \delta^2, \qquad r_2 = r^2, \\ v &= r r_2 - e_1(1 + s_1) r_2 + e_2(1 + s_1) r, \\ r_2 v &= (1 + s_1)(e_3 r_2 - e_4 r + e_5). \end{aligned} \tag{5}$$

These are four quadratic equations; the inverse weight is shared with the factor equations below.

Lemma 2.1. *The equations $q s_1 = 1$ and (5) imply*

$$(1 + q) P_5(r) = r^5, \tag{6}$$

with $P_5(r) \ne 0$. Conversely, any rational $r = \delta^2, q, s_1$ satisfying $q s_1 = 1$ and (6) has a unique extension to r_2, v satisfying (5). The case $r = 0$ is exactly $q = s_1 = -1$, $r_2 = v = 0$.

Proof. Put $N(r) = r^5 - P_5(r)$. Substituting the equations for r_2, v into the last equation of (5) gives

$$r^5 - (1 + s_1)N(r) = 0.$$

Multiplication by q, using $qs_1 = 1$, gives $qP_5(r) = N(r)$, which is (6). Reversing these steps proves the converse, and the displayed equations determine r_2, v uniquely. If $P_5(r) = 0$, then (6) forces $r = 0$, contrary to $P_5(0) = -e_5 \neq 0$. At $r = 0$, the last equation gives $1 + s_1 = 0$, and $qs_1 = 1$ gives $q = -1$. The other two outputs vanish. □

Since $P_5((t_* - 1)^2) = F_2(t_*)$, the lemma excludes every contact value. When $\delta \neq 0$, it gives precisely $R^{-1} = 1 + q$. The remaining stratum $\delta = 0$ has $q = -1$ and $M_2^\circ(t_*) = 1$; no value of R is used on that stratum. Every witness constructed in completeness has a selected contact prime with $v_p(q) = -1$, and therefore satisfies $q \neq 0$.

If both radicals divide a nonzero integer J to order k, then $h(T) \leq h(\lambda) + h(T\lambda)$ and (2)–(3) imply

$$B^2 \leq C|J|^{7/(6k)} + M_2, \qquad C = 2M_1H_2. \tag{7}$$

The two radicals need not be coprime for this implication.

3 A quadratic chart for the common reference point

Write $a_P = x_P$, $b_P = y_P$, and set

$$A_P = a_P^2 + D, \qquad B_P = a_P^2 - D, \qquad C_P = D^2 + 6a_P^2D - 3a_P^4.$$

The constants a_P, b_P, D, A_P, C_P are nonzero. The first four nonvanishing assertions follow from the displayed coordinates and $D = (3 - \rho)^2 > 0$ at both real embeddings; exact arithmetic gives

$$C_P = -\frac{94897379}{4096} + \frac{4195821}{256}\rho, \qquad \mathrm{Nm}(C_P) = -\frac{8203357319351}{16777216} \neq 0.$$

Retain four field coordinates S_H, T_H, f, F_0. Put

$$\begin{aligned}
L_f &= 2b_PD(a_PT_H - b_P),\\
Q_f &= a_PA_PS_HT_H + b_PA_PS_H - b_PB_PT_H + 4a_P^2D,\\
L_g &= 2b_PD(a_PS_H + b_P),\\
Q_g &= -a_PA_PS_HT_H - b_PB_PS_H + b_PA_PT_H - 4a_P^2D.
\end{aligned}$$

Impose the three quadratic equations

$$\begin{aligned}
L_f f &= Q_f,\\
L_g F_0 &= Q_g,\\
D^2fF_0 + a_PD(f + F_0) + a_PS_HT_H + b_P(S_H - T_H) + D &= 0.
\end{aligned} \tag{8}$$

Lemma 3.1. *These equations describe exactly the points $H \in E(K)$ for which H and $H - P$ are finite with nonzero abscissas, through*

$$S_H = \frac{y(H)}{x(H)}, \qquad T_H = \frac{y(H - P)}{x(H - P)}, \qquad f = \frac{1}{x(H)}, \qquad F_0 = \frac{1}{x(H - P)}.$$

In particular $fF_0 \neq 0$ at every solution. The statement includes all finite exceptional cases of the displayed equations.

Proof. For the algebraic calculation abbreviate $a = a_P$, $b = b_P$, $S = S_H$, $T = T_H$ and $g = F_0$. Thus $b^2 = a(a^2 - D)$. Define

$$\Phi = aS^2T^2 - (a^2 - D)(S^2 + T^2) + 2(a^2 + D)ST + 4aD$$

and the polynomial expressions

$$\begin{aligned} 2bDf_0 &= -(a^2 + D)S + (a^2 - D)T - bS^2 - aS^2T, \\ 2bDg_0 &= (a^2 + D)T - (a^2 - D)S - bT^2 + aST^2. \end{aligned}$$

Direct expansion gives

$$\begin{aligned} Df_0^2 + S^2f_0 - 1 &= \frac{a^2S^2 - b^2}{4ab^2D}\Phi, \\ Dg_0^2 + T^2g_0 - 1 &= \frac{a^2T^2 - b^2}{4ab^2D}\Phi, \\ L_ff_0 - Q_f = -a\Phi, &\qquad L_gg_0 - Q_g = a\Phi. \end{aligned} \tag{9}$$

Let $r_f = L_ff - Q_f$, $r_g = L_gg - Q_g$, and $R = D^2fg + aD(f + g) + aST + b(S - T) + D$. The further exact identity

$$\begin{aligned} L_fL_gR = D^2r_fr_g &+ (aDL_g + D^2Q_g)r_f \\ &+ (aDL_f + D^2Q_f)r_g - aD^2C_P\Phi \end{aligned} \tag{10}$$

shows that every solution has $\Phi = 0$, including the cases in which one leading coefficient vanishes. Also

$$(2bD)^2R(f_0, g_0) = D^2\big((a^2 - D) - aST - b(S - T)\big)\Phi.$$

If $L_fL_g \neq 0$, these identities give $f = f_0, g = g_0$. If $L_f = 0$, then $T = b/a$ and $\Phi = 0$ forces

$$S = \frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad L_g = -\frac{aDC_P}{a^2 + D} \neq 0.$$

Thus $g = g_0 = 1/a$. The coefficient of f in R is $D(a^2 + D)/a \neq 0$, so $f = f_0$. The case $L_g = 0$ is symmetric. In particular both leading coefficients cannot vanish simultaneously; at $S = -b/a, T = b/a$ one has $Q_f = C_P \neq 0$.

Equation (9) now proves $fg \neq 0$ and gives the points $H = (f^{-1}, S/f)$ and $H' = (g^{-1}, T/g)$ on E. Two more identities prove the translation:

$$a(f_0T + g_0S) + b(g_0 - f_0) = S + T,$$

$$f_0g_0(aS + b)^2 - a(1 - af_0)^2 = -\frac{(aS + b)^2\big(aST + b(S - T) + D\big)}{4b^2D^2}\Phi.$$

If $x(H) \neq a$, the second identity gives the abscissa of $H - P$ from the chord slope $(x(H)S + b)/(x(H) - a)$. The first states that $H, -H', -P$ are collinear, and therefore gives the required ordinate. If $x(H) = a$, then $S = \pm b/a$. At $S = b/a$ the formula gives $f = -a/D \neq 1/a$, which is impossible. At $S = -b/a$ it gives $H = -P$ and

$$T = -\frac{a^4 - 6Da^2 + D^2}{2b(a^2 + D)}, \qquad g = \frac{4b^2}{(a^2 + D)^2},$$

which are the coordinates of $H' = -2P$ in the stated form. Thus $H' = H - P$ in all cases.

Conversely, the chord-and-tangent formulas for any allowed H give $\Phi = 0$ and the expressions f_0, g_0 above; the same explicit doubling computation covers $H = -P$. The displayed identities then prove all three equations. This proves both directions. □

Write $P = 2R_0$ and $H = [m]R_0 + T_0$, with $T_0 \in E[2](K)$, and put $h = m/2$. This notation is an elliptic index; it does not define multiplication of a point by a nonintegral rational number.

4 The scalar certificate and the point domains

Put $\beta = 51X^2 + 2504$, let $S = Q_2 = [a]P$, and write $C_S = S - P = [a-1]P$. The scalar normalizer is

$$\begin{aligned} \ell &= 149, \qquad A_N = \ell^{2N+1}, \\ r_s &= \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \\ R_s &= \frac{A_N r_s}{1+\ell^2 r_s}. \end{aligned} \tag{11}$$

All five factors of r_s are positive, since $\beta \geq 2504$, so $R_s > 0$. The retained square-abscissa chart for S is

$$x_S = h_S^2, \qquad j_S^2 = x_S^2 - D, \qquad S = (x_S, h_S j_S).$$

It has $h_S x_S j_S \neq 0$. The fixed-subtraction coordinate $g_S \in K$ satisfies

$$8(x_P x_S + D)g_S + (39 - 36\rho)h_S + (8 - 14\rho)j_S = 0.$$

The rational component of g_S has a retained inverse witness. At $S = P$ the coordinate g_S is zero, so this guard gives $a \neq 1$. All nonzero points of $\mathbb{Z}P$ are nontorsion. Thus C_S is a nonzero nontorsion point, and the usual fixed-subtraction formulas have no remaining zero denominator. Explicitly, define

$$\begin{aligned} N_S &= (-660 + 417\rho)h_S + (-228 + 112\rho)j_S, \\ D_S &= (-704 + 384\rho)h_S + (660 - 417\rho)j_S + (6888 - 5040\rho)g_S. \end{aligned}$$

Then $D_S \neq 0$ and $z(C_S) = N_S/D_S$, where $z = -x/y$. These identities are obtained from the chord-and-tangent formulas; they are the same fixed-subtraction identities independently checked in the exact function field by the supplied certificate.

Use the common chart of Lemma 3.1, so $H = (f^{-1}, S_H/f)$ and $H - P = (F_0^{-1}, T_H/F_0)$. Before any logarithm is used, the scalar ratio coordinate $Z = Z_0 + \rho Z_1$ is determined by

$$ZD_S = -N_S S_H.$$

When H is nontorsion this is $Z = z(C_S)/z(H)$. Set $F = \mathrm{Nm}_{K/\mathbb{Q}}(f) \neq 0$. The reduced scalar norm equations are

$$A = \frac{Z_0^2 + R_s Z_1^2}{R_s^4}, \qquad W_s = \frac{A^2 + R_s F^2}{R_s^4}, \qquad b_s = R_s(R_s + 9792)W_s^2. \tag{12}$$

The actual quadratic source stores $R_s^2, R_s^4, Z_1^2, F^2, A, W_s$ and a scaled square of W_s; thus these equations are quadratic after clearing. The normalized scalar pole has exactly the ordinary pole input b_s in (12). To verify its scaling literally, let M be the coefficient of the stored W_s-square coordinate in source row491, let K be the coefficient of $e\bar{b}$ in row499, and let L be the coefficient of $\bar{b}p^2$ in row494. The exact coefficient identities are

$$K = 82479248M, \qquad 3L = 17K.$$

The rational substitutions

$$b_s = -K\bar{b}, \quad e' = -e/K, \quad p' = p/K, \quad u' = u/K, \quad v' = -v, \quad w' = w/K$$

convert those rows to the retained normalized pole equations. The independent checker verifies these identities. Since $R_s > 0$ and $F \neq 0$, one has $W_s > 0$ and $b_s > 0$ at every rational source solution.

Lemma 4.1. *Let $p > 76176$ or $p = 149$, and suppose $\nu = v_p(R_s) > 0$ is odd. Then*

$$v_p(F) > 5\nu/4, \qquad v_p(Z_0) > 23\nu/8, \qquad v_p(Z_1) > 19\nu/8.$$

Once the point interpretations hold, these imply

$$v_p(a-1) > 19\nu/8.$$

Proof. The number $9792 = 2^6 \cdot 3^2 \cdot 17$ is a unit at p. The valuation of the nonzero pole input is $\nu + 2v_p(W_s)$, so the pole obstruction makes it positive and $v_p(W_s) > -\nu/2$. The two nonzero terms of a norm $u^2 + R_s v^2$ have opposite valuation parity. A zero term can be omitted. Hence the second norm in (12) gives

$$v_p(A) > 7\nu/4, \qquad v_p(F) > 5\nu/4.$$

The first norm gives the two asserted bounds on Z_0, Z_1.

At some place above p the value $f(H)$ has positive valuation, because the valuation of its norm is positive. Thus H is in the formal group there. The displayed bounds give $v(Z) > 19\nu/8 > 0$, so $v(z(C_S)) > 0$. On $y^2 = x^3 - Dx$, at a good odd prime a point with positive z-valuation is either formal or reduces to $(0,0)$. Indeed $z^2 = x/(x^2 - D)$, and D is a unit. Translation by $(0,0)$ sends z to $-z$ and exchanges these two cases. Choose $C' = C_S + \epsilon(0,0)$ to be formal. The integral identity

$$[2m]C' = [4(a-1)]H$$

kills both two-torsion terms. The formal logarithm is an isometry, so its ratio has valuation $v_p((a-1)/h) = v(Z)$. Since $h = m/2$ is p-integral, $v_p(a-1) \geq v(Z) > 19\nu/8$. □

The integer $\ell = 149$ is prime. For the five contact roots $0, 1, 2401, -343/32, 1/2$, the residues of $(\theta - 2504)/51$ are respectively $59, 97, 109, 108, 78$ modulo 149. Their 74th powers are all 148, so they are nonsquares. Trial division through 12 certifies primality. The preceding lemma applies at 149: the curve has good reduction there, 9792 is a unit, and the pole obstruction excludes only $2, 3, 17$. The generic contact cutoff is unnecessary at this separately verified prime. Thus r_s is an ℓ-adic unit when X is integral at ℓ. If $v_\ell(X) < 0$, then $v_\ell(r_s) = 10v_\ell(X)$. It follows that

$$v_\ell(R_s) = \begin{cases} 2N+1, & v_\ell(X) \geq 0, \\ 2N-1, & v_\ell(X) < 0. \end{cases} \tag{13}$$

This is always positive and odd. These finite facts are included in the independent exact certificate.

Lemma 4.1, using only its norm inequalities at this stage, gives $v_\ell(F) > 0$. Every finite allowed point in

$$E[2](K) \ \cup \ (P + E[2](K)) \ \cup \ (-P + E[2](K))$$

has reciprocal-abscissa norm an ℓ-adic unit. To check this, use the two-torsion translations and

$$\mathrm{Nm}(x_P) = 41^2/64, \quad \mathrm{Nm}(x_P - (3-\rho)) = 17^2/64, \quad \mathrm{Nm}(x_P + (3-\rho)) = 63^2/64.$$

The remaining torsion abscissas have norm seven. Thus none of these points can equal H. In particular $h \notin \{-1, 0, 1\}$, H and $H - P$ are nontorsion, and all required denominators and formal-logarithm interpretations are now justified. This exclusion precedes every use of a comparison logarithm.

Let $c = |a - 1| > 0$. Outside the fixed cutoff, every prime of $M_5(\beta)$ is a positive odd contact of r_s and R_s. The contact at infinity contributes none, since a negative valuation of $\beta = 51X^2 + 2504$ is even there. Lemma 4.1 therefore implies $M_5(\beta)^3 \mid c$. The prime $\ell = 149$ is

below the defining cutoff of M_5, so it is automatically absent from $M_5(\beta)$. The forced-prime estimate is consequently independent of these radical factors. We obtain

$$\ell^{L_N} M_5(\beta)^3 \mid c, \qquad L_N = \left\lfloor \frac{19(2N-1)}{8} \right\rfloor + 1. \tag{14}$$

Together with (1), this gives

$$h(\beta) \le H \left(\frac{c}{\ell^{L_N}} \right)^{7/9} \le H \ell^{-L_N} c, \qquad c \ge \ell^{L_N}. \tag{15}$$

The last inequality uses $c/\ell^{L_N} \ge 1$. These scalar estimates are independent of the comparison logarithms.

5 Plain parameter ratios and the local tests

For $f = 1/x$ and $z = -x/y$, extend their regular values at the identity by zero. The retained point relations are

$$U_0 = H, \quad V_0 = H - P, \quad U_i = [u_i]P, \quad V_i = U_i - Q_i \qquad (1 \le i \le 4).$$

The charts permit $U_i = O$ or $V_i = O$ for $i > 0$. For each numerator pair retain its quadratic pair-chart coordinates $\mathcal{V}_i, \mathcal{Q}_i, \mathcal{W}_i, \mathcal{I}_i, \mathcal{K}_i$ as specified in [4]. Their exact relations give

$$z(U_i) = -\frac{\mathcal{V}_i}{\mathcal{I}_i}, \qquad z(V_i) = -\frac{\mathcal{Q}_i}{\mathcal{K}_i}, \qquad \mathcal{I}_i \mathcal{K}_i \ne 0,$$

including the identity branches with zero numerator. The common-point chart stores both negative reciprocal parameters $S_H = -1/z(H)$ and $T_H = -1/z(H-P)$. Thus the quadratic equations

$$L_i \mathcal{I}_i = \mathcal{V}_i S_H, \qquad L'_{i,0} \mathcal{K}_i = \mathcal{Q}_i T_H, \qquad L'_i = \sigma L'_{i,0}$$

mean precisely

$$L_i = \frac{z(U_i)}{z(H)}, \qquad L'_i = \sigma\Big(\frac{z(V_i)}{z(H-P)} \Big).$$

No numerator-parameter coordinate or logarithm correction is required. Put

$$D_H = f(H), \quad A_i = f(U_i), \quad B_i = \sigma f(V_i), \quad C_i = L_i - L'_i, \quad I = L_1 - \beta,$$

$$G = L_3^2 - L_2(L_2-2)L_3 L_4 + (L_2-2)^2 L_4^2 - 1.$$

The required orders are two for D_H, three for C_2, C_3, C_4, G, one for C_1, two for I, and one for the nine A_i, B_i. The final input comparison uses order one.

6 Five factors and extraction at one place

At a contact prime $p \notin \Sigma$, δ is a unit and $\nu = v_p(R) = v_p(F_2(t_*))$ is positive and odd, and $v_p(q) = -\nu$. This follows from the unique contact factor with positive odd valuation and does not require any condition on the valuation of the scale. All valuations of K are unramified and normalized by $v(p) = 1$. For a point on E, every finite value of $v(f)$ is even: if $v(x) < 0$ then $2v(y) = 3v(x)$, while if $v(x) > 0$ then $2v(y) = v(x)$; the remaining case is valuation zero. This also holds for conjugated reciprocal abscissas. The common-point lemma identifies $F_0 = f(H-P)$ on every solution, including the exceptional leading-coefficient cases. Its even valuation is therefore available before the exclusion of torsion classes and before any formal-parameter quotient is used.

Use

$$
\begin{aligned}
T_1 &= 1 + q^5 C_2^2 + q^2 D_H^2 + q^4 B_1^4,\\
T_2 &= 1 + q^5 G^2 + q^2 A_3^2 + q^4 B_0^4,\\
T_3 &= 1 + q^5 I^4 + q^3 A_2^2 + q^2 A_4^2 + q^4 B_3^4,\\
T_4 &= 1 + q^5 C_3^2 + q^2 C_1^4 + q^4 B_4^4,\\
T_5 &= 1 + q^5 C_4^2 + q^2 A_1^2 + q^4 B_2^4, \qquad W = T_1 T_2 T_3 T_4 T_5.
\end{aligned}
$$

Lemma 6.1. *At a contact prime, every factor is nonzero and has valuation at most zero. A factor of valuation zero is congruent to one in the residue field.*

Proof. First suppose every displayed test in a factor is nonzero. In each factor other than T_3, the three nonconstant valuations belong, respectively, to the odd integers, to $2 + 4\mathbb{Z}$, and to $4 + 8\mathbb{Z}$. For example T_1 has valuations

$$2v(C_2) - 5\nu, \qquad 2v(D_H) - 2\nu, \qquad 4v(B_1) - 4\nu.$$

The second has exact factor two because $v(D_H)$ is even and ν is odd. The third has exact factor four because $v(B_1)$ is even. The term $q^2 C_1^4$ in T_4 has valuation $2(2v(C_1) - \nu)$, again twice an odd integer.

In T_3 the four nonconstant valuations are

$$4v(I) - 5\nu, \quad 2v(A_2) - 3\nu, \quad 2v(A_4) - 2\nu, \quad 4v(B_3) - 4\nu.$$

The first three have residues $-\nu, \nu, 2$ modulo four. These are pairwise distinct and nonzero because ν is odd and the A_i have even valuations. The fourth has residue four modulo eight; it therefore differs from each of the first three and from zero. Thus, in every factor, all finite term valuations, including zero for the constant term, are distinct. Zero tests simply remove terms from this argument. The minimum is attained exactly once. This proves nonvanishing and the asserted bound. If the minimum is zero, every nonconstant term has positive valuation, so the residue is one. □

Proposition 6.2. *At each contact prime, the retained pole system and the cleared trace equations imply $\nu > 0$ and give all sixteen required test orders at one place above that prime.*

Proof. Write $W = w_0 + w_1\rho$. The literal cleared trace equations are

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad (1+q)\mathfrak{b} = \gamma^2\tau^2.$$

By Lemma 6.1, $W \neq 0$. The first equation therefore gives $w_0 \neq 0$ and $\tau = 1/\operatorname{Tr}(W^{-1}) \neq 0$; indeed $w_0 = 0$ would give $w_1 = 0$. Hence $\mathfrak{b} = \gamma^2 R/\operatorname{Tr}(W^{-1})^2$. The factor γ is a unit at every consumed contact prime. If $\ell = \max_{v|p} v(W) \leq 0$, the trace inequality gives $v_p(\mathfrak{b}) \leq \nu + 2\ell$. This valuation is odd; the retained pole obstruction forces it positive. Consequently $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, the factor valuations are nonpositive and add to ℓ. Each factor valuation is therefore at least ℓ, and each of its term valuations is strictly greater than $-\nu/2$.

The resulting lower bounds, before integer rounding, are

test	strict lower bound	integer order
C_2, C_3, C_4, G	$9\nu/4$	3
I	$9\nu/8$	2
C_1	$3\nu/8$	1
D_H	$3\nu/4$	2
A_2	$5\nu/4$	1
A_1, A_3, A_4	$3\nu/4$	1
B_0, B_1, B_2, B_3, B_4	$7\nu/8$	1

Here ν is a positive odd integer. The order two for D_H uses its even valuation. In fact each of the nine reciprocal-abscissa tests has order at least two; order one suffices for its subsequent formal-group use. Zero tests satisfy all the stated inequalities. □

7 Integrality before the formal congruences

At a contact place selected by Proposition 6.2, the points H, U_i are formal in the primary completion, and $H - P, V_i$ are formal in the conjugate completion. The reciprocal-abscissa tests have positive even valuation, so every nonidentity point has formal depth at least one. In the parameter $z = -x/y$, write

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) \equiv 1 \pmod{p^{4v(z(Q))}}.$$

The value of the unit is taken to be one at the identity. This is the constant-term formal logarithm estimate proved in [9].

The integral group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)](H - P)$$

kill the torsion in $H = [m]R_0 + T_0$. Their full logarithmic ratios are u_i/h and $(u_i - n_i)/(h-1)$, where $(n_1, n_2, n_3, n_4) = (n, a, t, b)$. At least one of $h, h-1$ is a p-adic unit, so at least one full ratio is integral. The formal units preserve the valuation of its plain parameter ratio. The positive-order test C_i transfers integrality to the other plain ratio, and then to the other full ratio.

Only after this transfer may the relative errors of order four be used as absolute congruences. They give

$$L_i \equiv u_i/h \pmod{p^4}, \qquad L_i' \equiv (u_i - n_i)/(h-1) \pmod{p^4}.$$

Since $n_i = h(u_i/h) - (h-1)((u_i - n_i)/(h-1))$, the high cross tests give $L_i \equiv n_i \pmod{p^3}$ for $i > 1$. The first cross test gives $L_1 \equiv n \pmod{p}$, while I gives $L_1 \equiv \beta \pmod{p}$. These arguments include identity numerator points, half-integral h, and inert primes, where conjugation preserves the unique valuation.

8 Global soundness for sufficiently large parameters

The scalar bounds (15) hold independently of the comparison logarithms. In particular $c \to \infty$ uniformly with N, and $B \geq c - 1$.

Let $\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$. The local comparison shows that each third power of a comparison radical divides Δ. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (7) gives

$$B^2 \leq C14^{7/18}B^{14/9} + M_2.$$

Since $14/9 < 2$, this bounds B by a fixed constant. Equation (15) gives $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$, a contradiction. Thus $\Delta = 0$ eventually. A stronger form of the integral Lucas descent gives

$$\log B \geq (c-5)\log(c-2) \qquad (c \geq 8). \tag{16}$$

Here is the complete elementary argument. Put $A_0 = |a|$ and $w = \operatorname{sgn}(a)(a-2)b$. Then $t^2 - A_0tw + w^2 = 1$. For $\eta = (A_0 + \sqrt{A_0^2 - 4})/2$, every positive norm-one unit of $\mathbb{Z}[\eta]$ is a power of η. Indeed multiply a unit by a power of η to place it in $[1, \eta)$; its coefficient of η is $(v - v^{-1})/(\eta - \eta^{-1})$, an integer in $[0, 1)$, so it is zero and $v = 1$. Let $U_0 = 0$, $U_1 = 1$, and $U_{j+1} = A_0U_j - U_{j-1}$ for $j \geq 1$. The unit identities show that $|w| = U_j$ and $|t| \in \{U_{j-1}, U_{j+1}\}$ for some $j \geq 1$. Modulo $A_0 - 2$ one has $U_j \equiv j$; modulo $A_0 + 2$ one has $U_j \equiv (-1)^{j-1}j$. Since

w is a nonzero multiple of $|a-2|$, it follows that $j \geq A_0 - 2$ if $a > 0$ and $j \geq A_0 + 2$ if $a < 0$. Induction gives $U_j \geq (A_0-1)^{j-1}$ for $j \geq 1$. Therefore $|t| \geq (A_0-1)^{A_0-4}$. Finally $A_0 \geq c-1$, which proves (16).

If $n - \beta \neq 0$, let J be its nonzero reduced numerator. The local input comparisons make both radicals divide J, while $|J| \leq 2B\exp(h(\beta))$. For $B^2 \geq 2M_2$, (7) gives

$$\log B \leq \frac{6}{5}\log(2C) + \frac{7}{5}\log 2 + \frac{7}{5}h(\beta) \leq C' + \frac{7}{5}H\ell^{-7L_N/9}c^{7/9}.$$

The last expression is $o(c)$ as $c \to \infty$, uniformly for $N \geq 1$. The Lucas bound $(c-5)\log(c-2)$ grows faster than c. Since $c \geq \ell^{L_N} \to \infty$ uniformly with N, these two inequalities are incompatible for all sufficiently large N. Hence $n = \beta \in \mathbb{Z}$. If $X = u/v$ is reduced with $v > 0$, then $v^2 \mid 51$, and squarefreeness of 51 gives $v = 1$. All bounds depend only on fixed constants. Since L_N is increasing, a single N_0 works for every larger N. Its dependence on the aggregate additive constant is ineffective.

9 Simultaneous completeness

Fix an integer input and $N \geq 1$. Choose a real neighborhood of a nonzero reference point $[h_*]P$, with $64 \mid h_*$, on which both values of $|z(H)|$ are bounded below. Then choose a sufficiently small real neighborhood of the identity for C_S. It gives $|Z_0|, |Z_1| \leq 1$. For nonzero $H \in \mathbb{Z}P$ the positive-square abscissa bound also gives $|\operatorname{Nm}(f(H))| \leq 1/7$. For $R_s \geq 1$, equations (12) imply

$$A \leq 2R_s^{-3}, \qquad W_s \leq 4R_s^{-10} + \frac{1}{49}R_s^{-3} < 5R_s^{-3}.$$

Therefore

$$0 < b_s \leq 25\frac{R_s + 9792}{R_s^5} < 17 \qquad (R_s \geq 10).$$

The displayed rational function is decreasing on $(0, \infty)$, and its value at ten is less than 17. Since $r_s \geq 1$,

$$R_s \geq \frac{149^{2N+1}}{1+149^2} \geq \frac{149^3}{1+149^2} > 10.$$

Thus the required real preparation holds for every $N \geq 1$ with explicit margins.

The finite preparation uses no new arithmetic theorem. For integer input, r_s is a positive integer and

$$R_s(R_s + 9792) = \frac{A_N r_s(A_N r_s + 9792(1+\ell^2 r_s))}{(1+\ell^2 r_s)^2}.$$

The numerator is coprime to the displayed denominator, so its reduced denominator is square. At the three preparation primes the elementary input congruences give

$$v_2(R_s) \in \{2, 3\}, \qquad v_3(R_s) = 1, \qquad v_{17}(R_s) = 0.$$

In particular $r_s \equiv 15 \pmod{17}$ and $1 + 149^2 r_s \equiv 3 \pmod{17}$, so the normalizer denominator is also a unit at 17. Thus $R_s(R_s + 9792) = R_s^2(1 + 9792/R_s)$ is a local square at all three primes. Choose the shared index h_0 sufficiently divisible at the finite support of this rational number, and then choose the index $c = a - 1$ still more divisible. The points $H = [h_0]P$ and $C_S = [c]P$ are then formal at the required finite places, and all components of $f(H)$ and Z have the needed arbitrarily large valuations. Make W_s integral at every prime where the displayed numerator has odd valuation, and deep enough at $2, 3, 17$ to supply the pole bounds. At every other prime the valuation of b_s is even. These conditions give a square reduced denominator and all pole preparation hypotheses. The source's progression and compact-real-group argument supplies the chosen real neighborhoods at the same time, together with the retained nonzero

chart guards. It also permits c to be arbitrarily large and to satisfy the retained divisibility conditions for the Lucas construction.

After S is fixed, continuity preserves this finite scalar preparation and its guards for all h in one fixed congruence class modulo a sufficiently divisible integer L. Retain the source's integral Lucas construction, choose the four root labels, and fix both positive scales and compact neighborhoods for all comparison tests before selecting contact primes. The additional contact primes avoid the finite scalar preparation support. Hence these later congruences do not conflict with any scalar congruence.

Choose a fixed positive integer D_0 clearing all contact coefficients $D_0^2\theta/a_*$ and sufficiently divisible at the prepared primes, including Σ, and the prime divisors of the scales. In the prime pattern write

$$\lambda = \frac{u}{D_0^2 w^2}.$$

The zero contact makes u one of the selected primes, outside that finite set. Thus $v_p(\lambda) \le -2v_p(D_0)$ there. Make this upper bound low enough that $t_* = a_*\lambda$ has valuation below zero and all nonzero contacts. Then $v_p(\delta) = v_p(t_*)$, and

$$R = \prod_{\theta\in\Theta_2} \left(1 + \frac{1-\theta}{\delta}\right) \longrightarrow 1 \tag{17}$$

in every prepared completion. The weight $q = R^{-1} - 1$ tends to zero. Increasing the fixed denominator divisibility makes its valuation uniformly arbitrarily large. Thus all weighted test terms are small at every prepared place, and $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two.

The factors in (17) are arbitrarily close to one. Requiring congruence to one modulo eight at 2, and modulo p at an odd prepared prime p, makes their product a local square. The pole input is $\gamma^2 R/\,\mathrm{Tr}(W^{-1})^2$, so it is also a square at $2, 3, 17$. The fixed factor has valuations

$$v_2(\gamma) = 3, \qquad v_3(\gamma) = v_{17}(\gamma) = 1.$$

Since the reciprocal trace is close to $1/2$, the pole input has valuations $4, 2, 2$ at those three primes. These satisfy all retained preparation inequalities.

This preparation is valid for any fixed nonzero rational scales. The prime-pattern input in [10, 1] applies to arbitrary fixed distinct integer coefficients. After clearing $D_0^2\theta/a_*$ and removing repeated coefficients, it applies here without a square restriction on a_*. All denominator conditions are fixed before prime selection.

Select the contact primes with the inherited opposite-order separation. Take $H = [h]P$, $U_i = [hn_i]P$ and $V_i = [(h-1)n_i]P$. The common-index congruences preserve the scalar progression and make H, U_i formal at one place and $H - P, V_i$ formal at its conjugate. Only natural depth one is needed on either branch; the source's stronger primary preparation is also permitted. Both full logarithmic ratios are exactly n_i. Plain-parameter errors therefore have order at least four. All C_i, and I since $n_1 = \beta$, have valuation at least four, and the Lucas equation for the chosen labels gives $v(G) \ge 4$. Every reciprocal-abscissa test has valuation at least two. At the selected place $v_p(q) = -1$, so every nonconstant term of every packed factor has positive valuation. Thus $W \equiv 1$ there. At the opposite place, each factor has valuation at most zero, with residue one when its valuation is zero. The trace of W^{-1} is a unit with residue one or two. The comparison pole input has valuation one at every selected contact prime.

The exact prime-pattern identity is

$$R = \frac{a_*^{10} \prod_{\theta\in\Theta_2} (u - (\theta/a_*)D_0^2 w^2)}{(D_0^2 w^2)^{10}\delta^{10}}.$$

Every factor outside the selected prime numerators is a rational square, including a_*^{10} even when a_* is not itself a square. Thus at every prime outside those numerators the valuation is even.

Multiplication by γ^2 and division by the squared trace preserve this parity. This argument includes primes dividing δ; no unit assertion about δ is made away from contacts. Equation (17) and the preparatory divisibility give the required local squareclasses and valuation bounds at $2, 3, 17$.

For the real preparation, write $r = \delta^2$ and let $\alpha_1, \dots, \alpha_5 > 0$ be the five squared distances of the contact pairs from one. Then

$$R = \prod_{j=1}^{5}(1 - \alpha_j/r), \qquad q \sim \frac{\alpha_1 + \cdots + \alpha_5}{r} > 0 \quad \text{as } r \longrightarrow +\infty.$$

On the fixed compact real neighborhoods the tests remain bounded, so $W \to 1$ and $\mathrm{Tr}(W^{-1}) \to 2$. Consequently

$$\mathfrak{b} \longrightarrow \frac{\gamma^2}{4} = \frac{41616}{167281} \in (0, 17).$$

All factor values are positive at both real embeddings for sufficiently large λ, because $q > 0$ and all test powers are even. The simultaneous selection supplies the required real size after all fixed data have been chosen. The pole-realization input supplies its witnesses, and the exact power identities fill the quadratic circuit. This proves completeness for every integer input and every $N \geq 1$.

10 Exact quadratic specification

Retain three scalar inverse weights s_1, s_2, s_4 for each detector, with the quadratic equations

$$qs_1 = 1, \qquad s_2 = s_1^2, \qquad s_4 = s_2^2. \tag{18}$$

Thus $q \neq 0$ and $s_j = q^{-j}$ for $j = 1, 2, 4$. The actual factor coordinates T_j are specified by the following quadratic equations:

$$\begin{aligned}
s_4T_1 &= s_4 + qC_2^2 + s_2D_H^2 + (B_1^2)^2,\\
s_4T_2 &= s_4 + qG^2 + s_2A_3^2 + (B_0^2)^2,\\
s_4T_3 &= s_4 + qI^4 + s_1A_2^2 + s_2A_4^2 + (B_3^2)^2,\\
s_4T_4 &= s_4 + qC_3^2 + s_2C_1^4 + (B_4^2)^2,\\
s_4T_5 &= s_4 + qC_4^2 + s_2A_1^2 + (B_2^2)^2.
\end{aligned} \tag{19}$$

Every test square on the right is a stored field value. Squaring the stored B_i^2 is quadratic in its two rational components and needs no output for B_i^4. Multiplication by q^4 proves exact equivalence with the five factor definitions above. The products of the factors and the pole normalization are unchanged by this clearing.

The three contact outputs r, r_2, v are specified by (5), using the already retained inverse weight s_1. Their equations involve only products of two stored quantities. No coordinate for $P_5(r)$ is required. Lemma 2.1 proves the exact rational interpretation and treats the $r = 0$ stratum without division by r. The scalar normalizer is the quadratic equation $R_s + 22201 r_s R_s - A_N r_s = 0$. The scalar chain is $R_2 = R_s^2$, $R_4 = R_2^2$. The two scalar norm rows use R_4, as in (12).

Only C_1^4 and I^4 are retained as separate field fourth powers. The five B_i^4 values occur directly as squares of stored field coordinates in (19). All remaining point and scalar equations

are fixed by the literal coefficient tables. The exact count is

point, scalar, contact, ratio, and weight coordinates	190
two shared fourth powers	4
five factors and intermediate products for two detectors	32
total coordinates, including the input	226
point, scalar, contact, ratio, and weight quadratic rows	160
shared fourth-power rows	4
factor and product rows	36
total quadratic rows	200

The complete tables fix every coordinate, equation, coefficient, and normalization; the input is coordinate one. For the common point, the six primitive rational component equations are replaced by the fixed rational row combinations in the certificate, whose determinant is $-1/362963666515968$. The producer clears denominators and makes each resulting row primitive before squaring. The row change is nonsingular and preserves the common zero set. The independent coefficient checker imports no producer. It computes the elementary symmetric sums of the five squared contact distances, checks the five roots of P_5, and verifies the general polynomial identity in Lemma 2.1. It also verifies the original normalization algebraically after multiplication by q; the discrepancy is exactly a multiple of $qs_1 - 1$. The nonzero constant term of P_5 certifies the exceptional-stratum argument. The checker inspects all occurrences of every omitted polynomial coordinate in the pinned source. It reconstructs the 4 changed contact rows and compares the other 196 rows exactly with that source. It independently expands all 200 quadratic squares and checks every coefficient, every coordinate, the degree, the content, and the coefficient-height formula.

The reconstruction dependencies retain separate exact certificates for the other algebraic components. The common-point certificate verifies twelve polynomial identities, including elimination without division and translation on every exceptional stratum. The factor certificates check the residue classes for all odd contact orders, the resulting comparison precisions, and the completeness depths. The contact-set certificate checks the exceptional primes and the height-transformation matrices. The scalar certificate checks the forced prime, its nonsquare residue tests, and the extraction bounds. These finite certificates support the algebraic specification; the infinite arithmetic arguments and their premises are the proofs above.

The expanded quartic has 2836 terms before powers of A_N belonging to the same coordinate monomial are collected, and 2835 monomials after collection. Its unique coefficient involving A_N^2 is $A_N^2 + 1$, at the square of the actual scalar contact-product coordinate. Every other coefficient is a nonzero constant or a nonzero constant multiple of A_N. Consequently no support cancellation occurs for positive A_N. The largest constant coefficient C_0 has 1520 bits, the largest linear multiplier is 44402, and exact comparison gives

$$C_0 > 44402^2, \qquad 149^{210} + 1 < C_0 < 149^{214} + 1.$$

Thus $H(P_N) = \max(C_0, A_N^2 + 1)$ for every $N \geq 1$ and the asserted exact height holds for $N \geq 53$. This is a coefficient statement; it does not make the defining threshold effective. The producer `build_inverse_contact_v21.py` and the independent checker `check_inverse_contact_v21.py` reconstruct all 200 rows and every coefficient. The full coefficient tables and certificate are in `computation/quartic/family226_v21`.

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